CENTRAL MANUFACTURING TECHNOLOGY INSTITUTE

C – Smart Manufacturing Cell

User Guide

Troubleshooting – M2M copy and connectivity

PROJECT: SPV Data Flow

This guide covers the diagnosis and resolution of connectivity and copy operation issues in SPVDataFlow application.

How Connectivity Is Evaluated

The application uses two distinct methods to determine machine health. Understanding the difference is essential for accurate diagnosis:

- **Quick Reachability Check** — The server attempts to open TCP port 445 (SMB) on the configured IP address. This test confirms basic network reachability only. It does not validate UNC paths, share names, or credentials.
- **Real Copy Connection** — A full SMB session is established to the UNC path using the stored credentials. Failures here appear as connection_failed or copy_failed in logs and history, even when port 445 is reachable.

UNC Path — Format Reference & Examples

The Network Path field is the most common source of copy failures. The path must be a valid Windows UNC path in the exact format below.

One wrong character causes connection_failed even when port 445 is open.

```
\\<host-IP-or-hostname>\<share-name>\<subfolder>\<subfolder2>
// Example — IP-based:   \\192.168.1.50\DataShare\EL\LINE01
```

Correct vs. Incorrect — Common Mistakes

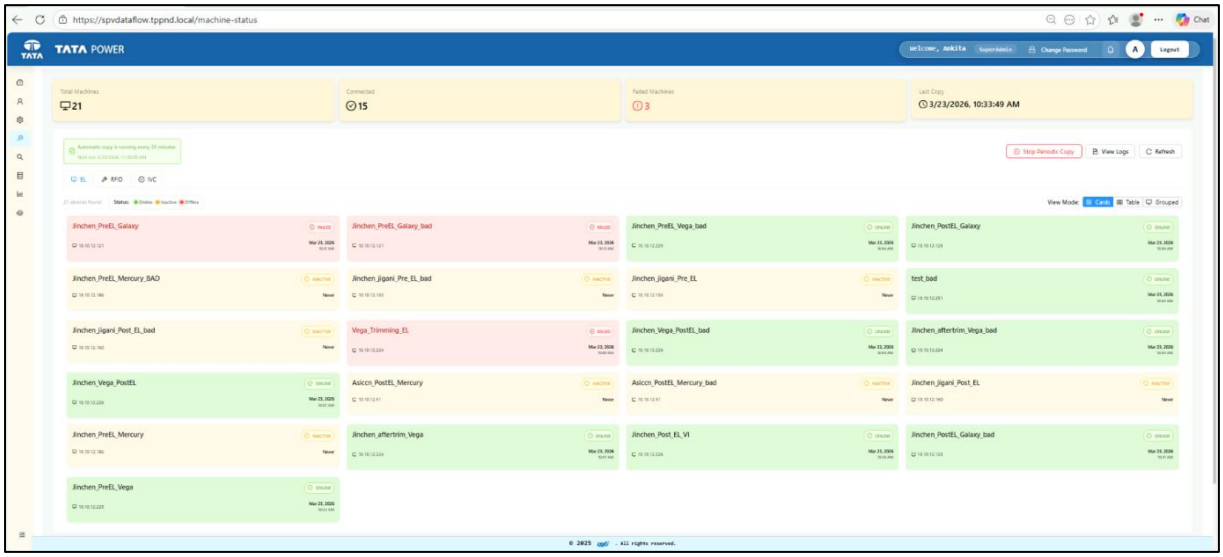
Scenario	Correct UNC Path	Common Mistake
Basic share	`\\192.168.1.50\DataShare`	`\\192.168.1.50/DataShare` (forward slashes)
Subfolder path	`\\192.168.1.50\DataShare\Files`	`\\192.168.1.50\DataShare\Files\` (trailing slash)
IP + subfolder	`\\10.0.5.22\Share\RFID\GATE_A`	`\\10.0.5.22\Share/RFID/GATE_A` (mixed slashes)
Custom dest path	`\\172.18.100.67\CustomPath\IVC`	`\\172.18.100.67\CustomPath` (missing type subfolder)

Verify manually first: Before saving to the app, open Windows Explorer (run) on a PC that is on the same network as the application server and type the UNC path directly into the address bar using the same credentials. If Explorer cannot open it, the portal copy will fail too.

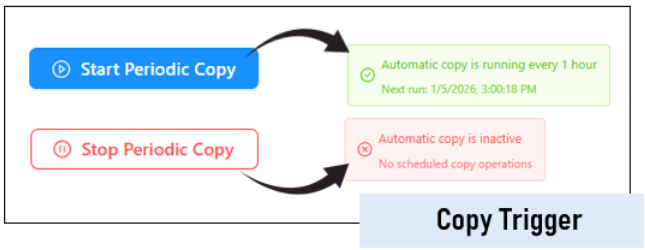
Understanding Status Labels

The Machine Status screen displays one of the following labels for each registered device. Use this table as a quick reference before diving into detailed troubleshooting.

Status Label	Typical Meaning	First Checks	Severity
ONLINE	Port 445 reachable; machine is active	Check UNC path and credentials if copy still fails	**Low**
OFFLINE	Port 445 check failed or data missing	Verify power, IP address, firewall, SMB service	**High**
INACTIVE	Machine disabled in configuration	Re-enable in Machine Configure	**Medium**
FAILED	Latest copy/connection operation failed	Open Copy Logs and read the error_message field	**High**



Please ensure that the “Start Periodic Copy” is enabled to ensure data is being copied once in every 30 mins.



Section 1: Machine is OFFLINE

What OFFLINE means: The application server could not open TCP port 445 on the configured IP address. Run these commands FROM THE APPLICATION SERVER (10.10.12.73 or a machine on the same subnet as the server), not from your workstation.

1.1 Step-by-Step Diagnosis — Run in Order

Step	Action / Command	Expected Result
1	Confirm the device is powered on. Check the physical machine or ask the floor operator.	Device is on and is ONLINE in the machine status page
2	Ping the configured IP from the application server to confirm basic reachability. `ping <source_ip>`	Replies received, 0% loss. If all timeout, go to step 3.
3	Test port 445 (SMB) directly. This mirrors exactly what the application does. `Test -NetConnection -ComputerName <source-ip> -Port 445`	TcpTestSucceeded : True. If False, check firewall (step 5).
4	If ping fails, trace the route to identify where packets drop. `tracert <source-ip>`	Identifies the hop where traffic is lost (router/firewall).
5	Check Windows Firewall on the target device (run on the device itself, not the server). `netsh advfirewall firewall show rule name="File and Printer Sharing (SMB-In)"`	Rule exists and Enabled=Yes. If missing, add it (step 6).
6	Add the SMB inbound firewall rule on the target device if missing. `netsh advfirewall firewall add rule name="SMB-In" protocol=TCP dir=in localport=445 action=allow`	Rule added. Retest port 445 from the server.
7	After all fixes, click Refresh on Machine Status. If still OFFLINE after 60 seconds, re-run Test-NetConnection to confirm port 445 is now open.	TcpTestSucceeded : True — machine should now show ONLINE.

Step	Action / Command	Expected Result
	`Test-NetConnection -ComputerName <source-ip> -Port 445`	

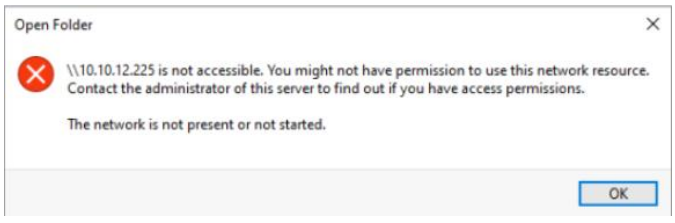
If the issue still persists, contact the IT team of TPREL to troubleshoot network and firewall issues.

Section 2 : Machine is ONLINE but Status is FAILED

What FAILED means here: Port 445 is open (machine shows ONLINE briefly) but the full SMB connection using credentials and UNC path fails. Error appears as connection_failed or copy_failed in Copy Logs. This is a path/credentials/permission issue, not a cable or power issue.

Always open Copy Logs and read the error_message field before running any commands. The error text tells you exactly which sub-type of failure occurred:

Error Message Contains	Root Cause & Go To
`Logon failure / Access is denied`	Wrong username or password , Contact TPREL IT team to give correct credentials
`The network path was not found`	UNC path is incorrect or share does not exist
`Multiple connections to a server`	Conflicting SMB sessions on the server, Close any current sessions which are present (like opened shared folders) and retry again.
`Windows network APIs not available`	Source machine is not windows based
`The password has expired`	Domain password policy expired the credential , Ask TPREL IT team to share updated password and fix in machine configure page.



If error messages like these are encountered, please restart the backend services of the application and try again
If issue persists , contact TPREL IT team for troubleshooting

In detail error codes , meaning and possible resolutions

Path / name / reachability

Code	Typical message	Common cause	What to do
1203	Network path typed incorrectly, does not exist, or provider not available	Wrong **UNC** (\\server\share\...), share removed/renamed, **offline** host, wrong **DNS** name	Restart EL_BACKEND service Fix **UNC** in Machine configure; open same path in Explorer from a PC on the same network as the **app server**; verify share exists and machine is on.
53	Network path was not found	**Host not found** (name resolution), wrong computer name, IP changed	Ping/nslookup from **server**; use **IP** in UNC if DNS is bad;
67	Network name cannot be found	Bad share name, name typo	Correct \\server\ShareName; ensure **File and Printer Sharing** and share name match
64	The specified network name is no longer available	Server went **offline**, SMB dropped,	Stabilize host; retry; check SMB service and network
1231	Network location cannot be reached	**Routing/VLAN/VPN**, firewall path, no L3 path to subnet	IT: route/firewall from **application server** to device network

Credentials / account

Code	Typical message	Common cause	What to do
1326	Logon failure: unknown user name or bad password	Wrong user/password, typo, **expired** password	Update credentials in configure;
86	The specified network password is not correct	Same as bad password (older wording)	Same as 1326
1909	The referenced account is currently locked out	Too many bad attempts	IT: unlock account; fix password
1331	Logon failure: account currently disabled	Account disabled in AD	IT: enable account
1790	The account is expired	AD account expired	IT: renew account

Session / “already connected” conflicts

Code	Typical message	Common cause	What to do
1219	Multiple connections to a server… different user	Same server already mapped **as another user**	Disconnect existing mappings (net use) on the **app server**; use one consistent account; or connect with same user
85	Local device name is already in use	Drive letter conflict (if mapping used letters)	Usually N/A if using UNC-only; else free the device

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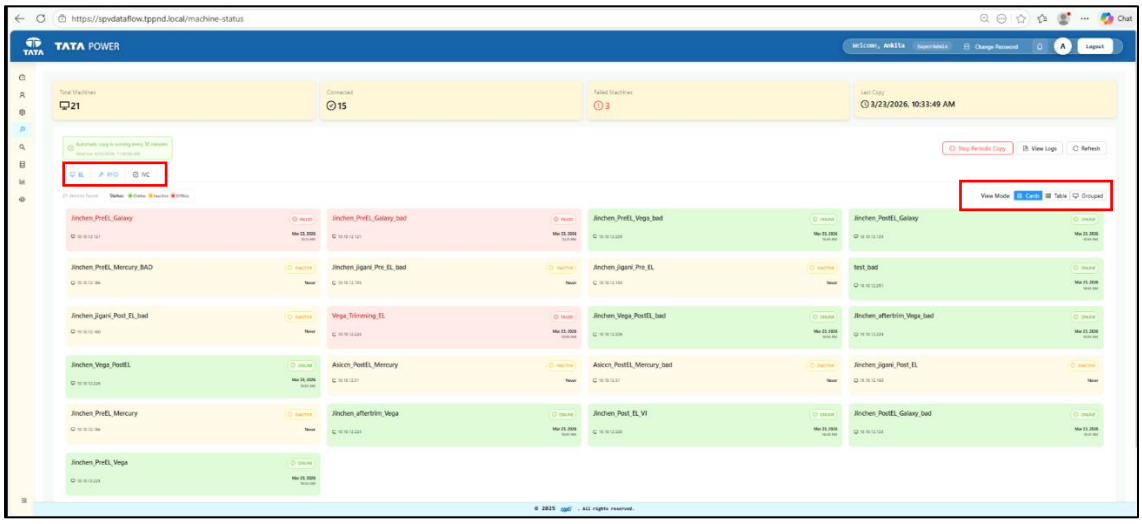
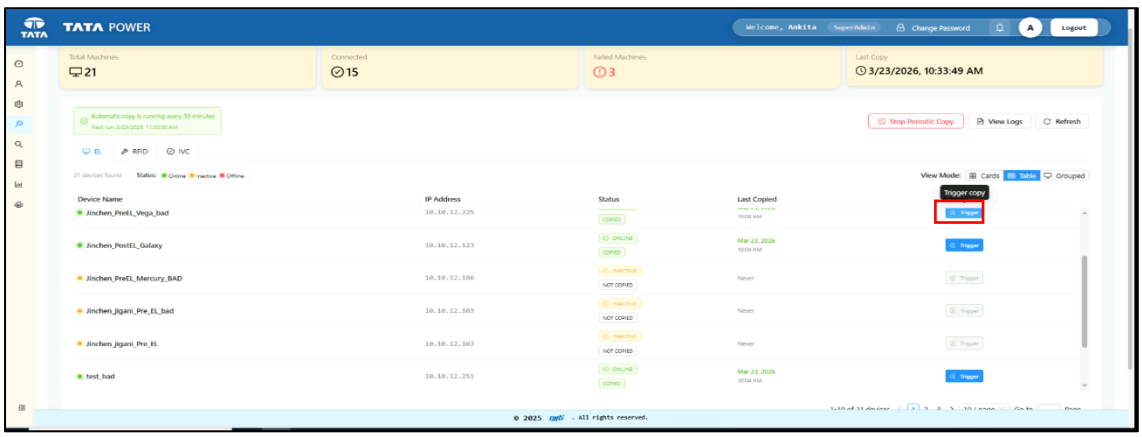
Technical note: Port 445 open (ONLINE) ≠ Copy will succeed. Always verify UNC path, credentials, and share permissions separately

Section 3 : Steps to follow once the machine copy issue is resolved

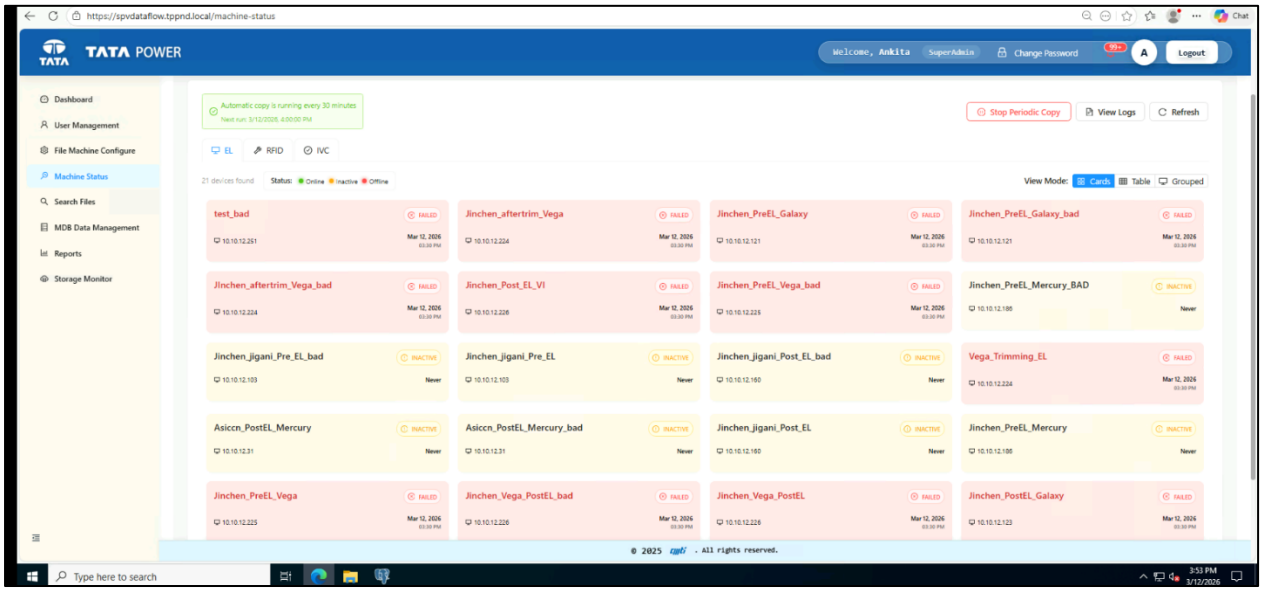
Step 1 : Navigate to the Machine Status Page , Click on the machine type tab (EL/ RFID / IVC)



Step 2 : Click on the table mode , Manually trigger to check if data is being copied , if a Success message appears, it indicates that the issues is resolved.



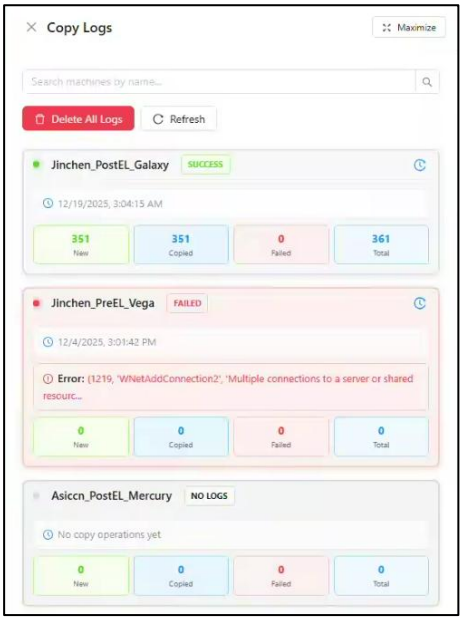
If all the machines show failed copy.



- Immediately check the copy logs (View logs button in the machine status page) to confirm when the copy was stopped and the reason of failure, additionally you can check the folder ‘F:/2026_FILE_REPO’ on 10.10.12.73 to verify when copy was stopped.

  Restart the EL_BACKEND in services and check again.

- In the windows explorer, type the unc path of a machine and confirm if it is opening on the server, refer section 2 based on error that was displayed.

- If issue still persists, contact TPREL IT team for the same to check for network changes, firewall changes etc.

  Once the issue has been resolved , follow section 3.